

# Choosing optimal $\varepsilon$ under privacy-vulnerability correlation

Vihaan Gupta\*, Jade Nair†, Julia Shephard‡, Natalia Siwek§

June 27, 2026

## Abstract

A large literature in differential privacy focuses on optimizing parameters like  $\varepsilon$  in light of the privacy-utility tradeoff. We start with the observation that individuals whose data is often most useful for research studies (such as those with rare diseases or gender minorities) tend to have the strongest preferences for privacy. We then construct a model to determine the optimal  $\varepsilon$  for a study in such a setting, where individual preferences for privacy and vulnerability are highly correlated. After using simulations to explore the implications of our model, we present a proof-of-concept method for debiasing estimators produced without the participation of high-privacy high-vulnerability individuals. Finally, we construct a tool to allow individuals to determine their preferred  $\varepsilon$ , to help us understand heterogeneous distributions of privacy preferences in the real world.

## 1 Introduction

Decision-making based on sensitive data is increasingly desirable in an information-rich world. Medical studies that collect participants’ health histories facilitate cures for diseases, targeted advertising helps individuals to purchase products that they want, and social science studies enable more equitable redistricting and division of government resources. These cases motivate the development of privacy-enhancing technologies, tools that allow us to draw useful insights from data while respecting individuals’ privacy and rights.

**Differential Privacy:** One particularly popular technology is Differential Privacy (DP). The goal of DP is to ensure that, with high probability, we cannot distinguish between two datasets that are the same save one person’s membership in the dataset. More formally, let us define two neighboring datasets

$$x \sim x'.$$

We say that  $x$  *neighbors*  $x'$ ; that is,  $x$  and  $x'$  are identical datasets save for whether one individual’s data is present or not in them. Differential privacy constructs mechanisms  $M(x)$  for fixed privacy parameter  $\varepsilon > 0$  such that for any set  $E$ ,

$$P(M(x) \in E) \leq e^\varepsilon P(M(x') \in E).$$

---

\*vihaangupta[at]college[dot]harvard[dot]edu. Conducted simulations, assisted with development of the model, contributed heavily to writing of theory sections.

†jadenair[at]college[dot]harvard[dot]edu. Led development of the technical tool and corresponding writeup, assisted with the model.

‡jshephard[at]college[dot]harvard[dot]edu. Led theoretical work on the model and the bias adjustment section; contributed heavily to introduction and conclusion writing, as well as writing of theory sections.

§nataliasiwiek[at]college[dot]harvard[dot]edu. Conducted simulations, contributed heavily to introduction and conclusion writing, assisted with the model.

In global differential privacy, the parameter  $\varepsilon$  is set by a trusted curator and manages the tradeoff between privacy and data utility. With a sufficiently low  $\varepsilon$ ,  $M(x)$  will output a differentially private statistic, such as a noised mean or a differentially private regression, that cannot be distinguished from  $M(x')$ . Dwork 2011<sup>1</sup>

Differential privacy presents a natural tradeoff: how to choose an optimal  $\varepsilon$  to balance privacy and utility goals. On the one hand, a higher chosen  $\varepsilon$  produces more accurate statistics, on the other, a lower  $\varepsilon$  protects the identity of study participants better. This question is complicated by the fact that individuals often have different preferences for privacy and that those with the most valuable (or sensitive) data, such as rare disease patients, often have the strongest preferences for privacy.

Work from both computer science and economics has tried to deal with appropriately evaluating this tradeoff. We know of two sets of literature that do this. The first set of papers presents mechanisms to differentially compensate participants for their privacy loss—individuals are paid such that the benefits of participating in the study outweigh the costs. Ghosh and Roth 2010 sets up such a method in the context of auctions, while Krehbiel 2019 develops a private mechanism such that individuals may pay to avoid having their data used. The second set of papers models the privacy-utility tradeoff in specific contexts without considering losses to utility when the strictly most vulnerable people participate. An early model of this is Hsu et al. 2014, which outlines privacy-utility tradeoffs in five different use cases. The Census Bureau’s implementation of differential privacy seeks to maintain a given level of privacy protection and improve utility at that privacy level Abowd et al. 2022. Meanwhile, Lin and Xiao 2024 simultaneously optimizes for privacy and utility but does so in a population where privacy preference and vulnerability to data release are not highly correlated.

Our project extends existing work on determining optimal  $\varepsilon$  values by excluding the correlation between privacy preferences and vulnerability level. We do so while departing from the payment mechanism designs tradition. Disclosing desired compensation can itself breach privacy, as it may signal the sensitivity or vulnerability of an individual’s data—for instance, if the data relates to specific medical conditions. In such cases, even the act of opting out could reveal private information. Further, we expect that not all pieces of data are equally valuable. In particular, the failure mode in which individuals with the most sensitive data all refuse to participate in a given study, is salient and alarming. We think this work is especially important in contexts such as healthcare, justice, or housing, where marginalized groups are often underrepresented in research.

We present a theoretical model to choose  $\varepsilon^*$  (the optimal  $\varepsilon$  for a given study, hereafter referred to as  $\varepsilon$ ) based on the choice of  $\varepsilon$  that study participants would most prefer. To ensure that individuals do not reveal information about their vulnerability by openly choosing not to participate, individual preferences are not disclosed; instead, researchers access data only from individuals whose privacy preference ( $\varepsilon$  threshold) is less strict than the chosen research  $\varepsilon$ . Throughout our model, we assume that researchers are interested in statistics related to individual vulnerability (e.g. their experiences with a rare disease, crimes they have committed, or sexual behaviors), and that those with more sensitive answers have stronger privacy preferences (i.e. for higher  $\varepsilon$ ). Since we do not use differential compensation to encourage more vulnerable individuals to participate, we assume *plausible deniability*: individuals who do not participate in a given study can attribute it to a lack of interest that is evenly distributed across different vulnerability levels. In the setting of a medical study, this

---

<sup>1</sup>An alternative approach would be to use local differential privacy such that individuals could add noise according to their personal preferences. We think that this is difficult, however, as it would be important for a central curator to understand how much noise was added at each step in order to produce and interpret statistically meaningful results. And this presents the same challenge stemming from the correlation between preferences for privacy and vulnerability to data release that we outline below.

seems reasonable to us - the vast majority of opt-outs are not due to privacy concerns.

## 2 Finding optimal epsilon

We model the setting of a research study in which each individual  $i$  has some characteristic  $X_i \in [-b, b]$ . Let  $f : [-b, b]^N \rightarrow \mathbb{R}$  represent a function that takes individuals' data and outputs a statistic of interest  $S$ . We have that  $M(f(x))$  releases a differentially private statistic  $\tilde{S}$ , where  $\tilde{S} = S + \text{Lap}(s/\epsilon)$ , with sensitivity  $s$ .

### 2.1 Modeling privacy

We assume that individuals have a vulnerability score  $V_i$ , which captures how much risk they face from data release. This vulnerability is a function of the characteristic  $X$ , i.e.,  $V_i = g(X_i)$  for some increasing function  $g$ . For simplicity, we take  $V_i = |X_i|$  in our base model. We assume that individuals' preferences for privacy are heterogeneous and correlated with the costs an individual would face in the event of data leakage or data release. More specifically, we write that for every individual  $i$  in our population,

$$\epsilon_i = \beta(1/V_i) + u_i; \quad (1)$$

Here,  $\epsilon_i$  represents the maximum  $\epsilon$  at which an individual  $i$  would be willing to participate in a study. This "personal epsilon" depends on individual preference and is inversely proportional to the individual vulnerability  $V_i$  with some constant factor added. Those facing the greatest data risk (e.g. those who are a gender minority, have broken the law, have a rare disease) often, but not always, have the strongest preferences for privacy. Our  $u_i$  coefficient represents an individual's personal proclivity to participate in a research study, independent of one's vulnerability. Individuals who do not participate can use a dislike of participation to explain lack of entry into a research study. We can adapt our functional form such that the relationship between preferences for  $\epsilon$  and vulnerability  $V_i$  is quadratic, exponential, or related in some other way; we focus on the linear case for simplicity.

The privacy of individual  $i$  resulting from a lack or surplus of privacy is

$$U_{i;\text{privacy}} = \begin{cases} 0 & \text{if not participate} \\ \epsilon_i - \epsilon & \text{if participate} \end{cases} \quad (2)$$

Throughout the paper, we will reference to an individual's personal epsilon preference as  $\epsilon_i$  whereas the *chosen* epsilon for the model set by the curators as  $\epsilon$ . While one is relative to individual preferences, the other is the final optimal value chosen for the model based on individual preferences. Thus, if an experiment has privacy budget  $\epsilon$ , individuals with privacy value  $\epsilon_i < \epsilon$  will not participate, while individuals with  $\epsilon \geq \epsilon_i$  will participate and will have an additional privacy factor  $\epsilon_i - \epsilon$  taken into account. The personal privacy benefit for a single individual who opts in can be modeled with the following term

$$\frac{\max(\epsilon_i - \epsilon, 0)}{\epsilon} \quad (3)$$

## 2.2 Modeling utility

We consider societal utility as a result of accurate research studies. We follow the methodology in Lin and Xiao 2024 in noting that utility decreases with noise; that is, writing

$$U_{data} = 1 - \frac{1}{n\varepsilon} \quad (4)$$

where  $n$  is the number of participants who opt in for our data study. We calculate  $n$  using the formula

$$n = N \cdot (1 - F(\varepsilon)) \quad (5)$$

since  $(1 - F(\varepsilon))$  is the proportion of population where  $\varepsilon_i \geq \varepsilon$  where  $F$  is the CDF of the distribution of personal epsilons that we assume. While we are also concerned with the bias in our study, we do not incorporate a term involving bias into our data utility, since we cannot a priori know how much bias has been added. Since individuals independently derive utility from the quality of the research study, we write cumulative utility from data as

$$U_{all;data} = N(1 - \frac{1}{n\varepsilon}) \quad (6)$$

## 2.3 Solving for social welfare

To calculate the  $\varepsilon$  that is optimal for social welfare, i.e. balances both privacy and utility, we combine our utility functions for both data and the privacy of individuals. We assume that the range of possible  $\varepsilon$  our task might take on are distributed over a plausible range. Our initial assumption is that  $\varepsilon_i \sim \text{Unif}(\varepsilon_a, \varepsilon_b)$ ; for example,  $\varepsilon_i$  could be a real number drawn with uniform probability. However, our method is not dependent on this assumption, and we perform simulations that do not require it. We seek to maximize social welfare, which we define as the sum of the welfare resulting from utility derived from a good study (2.2) and the welfare resulting from privacy for individuals who participate in the study (2.1).

We calculate the former using formula (6) but take two things into account. For one, the utility that an individual derives from a study is directly proportional to their vulnerability  $V_i$  since more vulnerable individuals are often the ones who need the study the most. We take this into account by multiplying (6) with the term  $1/\varepsilon_i$  since we assume a converse linear relationship between vulnerability and personal epsilon. Secondly, the study has a higher utility based on the number of vulnerable individuals who opt in. We take this into account by taking a cumulation of  $1/\varepsilon_i$  but only for those individuals for whom  $\varepsilon_i \geq \varepsilon$ , i.e they participate in the study.

Now, for the second term that takes into account the welfare of the study based on the privacy it guarantees, we simply add a cumulation of individual privacy individuals participating in the study achieve given by (3).

Adding both these factors, we get the social welfare formula:

$$W = N \int_{\varepsilon_a}^{\varepsilon_b} \left( \frac{1}{x} \right) \left( 1 - \frac{1}{n\varepsilon} \right) \int_{\max(\varepsilon, \varepsilon_a)}^{\varepsilon_b} \frac{1}{y} dy dx + \int_{\max(\varepsilon, \varepsilon_a)}^{\varepsilon_b} \left( 1 - \frac{\varepsilon}{x} \right) dx \quad (7)$$

Those terms in order respond to utility from data different for individuals with different levels of vulnerability and then the additional privacy factor.

We simulate the welfare functions for varying bounds  $\varepsilon_a - \varepsilon_b$  and the number of participants  $N$  to get the optimal epsilon for each bound. We conduct different simulations for different assumptions of personal epsilons: uniform, normal and mixture of normals.

## 2.4 Simulation results

To perform the simulations we use the formula that

$$\mathbb{E}[f(x)] = \int_a^b f(x)p(x) dx \quad (8)$$

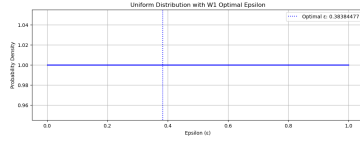
where  $p(x)$  is the pdf. This means we need to incorporate the pdf into the integral calculations. We list parameters in Footnote 2.<sup>2</sup> We want to explore 5 distributions as potential preferences for the users in the population - uniform, normal, mixture of normals, exponential and reversed exponential.<sup>3</sup>

---

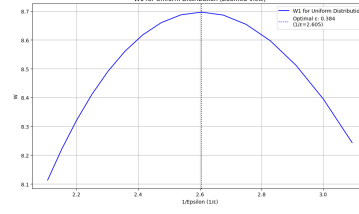
<sup>2</sup>We set the following values:

$$n = 5.0, N = 100, \varepsilon_a = 10^{-8}, \varepsilon_b = 1.$$

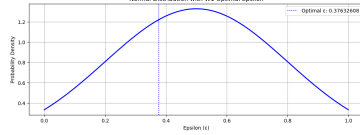
<sup>3</sup>The Uniform has bounds as  $\varepsilon_a$  and  $\varepsilon_b$ , The Normal is distributed as  $\mathcal{N}(0.5, 0.3)$ , Mixture of Normals is distributed as  $0.6\mathcal{N}(0.3, 0.1) + 0.4\mathcal{N}(0.6, 0.2)$ , Exponential has parameter of 0.5, Reversed exponential has parameter of 0.5



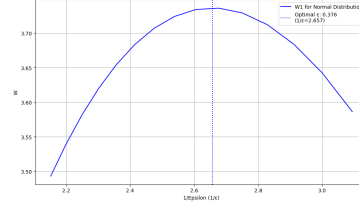
(a) Uniform: Epsilon Distribution



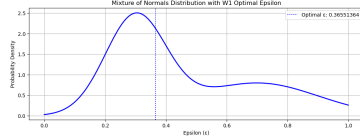
(b) Uniform: Welfare Function



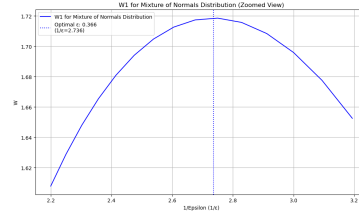
(c) Normal: Epsilon Distribution



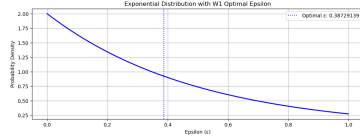
(d) Normal: Welfare Function



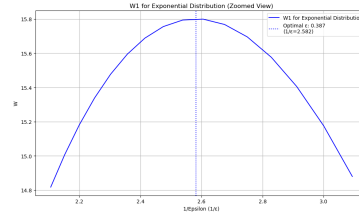
(e) Mixture: Epsilon Distribution



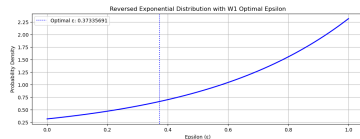
(f) Mixture: Welfare Function



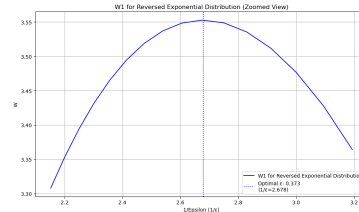
(g) Exponential: Epsilon Distribution



(h) Exponential: Welfare Function



(i) Reversed Exp: Epsilon Distribution



(j) Reversed Exp: Welfare Function

Figure 1: Epsilon distributions, corresponding welfare functions under different population models.

We get the following results for distributions - for uniform:  $\varepsilon = 0.384$ , for normal:  $\varepsilon = 0.376$ , for mixture of normals:  $\varepsilon = 0.366$ , for exponential:  $\varepsilon = 0.387$ , for reversed exponential:  $\varepsilon = 0.373$ . We can note that the results are in the similar range, with small differences due to specific distribution. The developed model appears to capture changes in user preferences and the underlying distribution within the domain. These results highlight the model's ability to adapt to varying privacy preference profiles, balancing the utility gained from user inclusion against the cost of added noise. They also emphasize the importance of accurately modeling user privacy preferences. We can note that all distributions have fairly similar preferred epsilon. It should be noted that they use the same

constants within the welfare function and bounds on  $\varepsilon_a$  and  $\varepsilon_b$  and that their means are also reasonably close to each other. However, the differences between the results highlight the ability of the welfare function to capture the differences in the distributions and pick an appropriate optimum epsilon that maximizes the social welfare for a specific distribution.

We can note a few limitations to the presented simulations. First, the optimal epsilon is impacted by the choice of constant factors, which can be fairly arbitrary but might also be relevant—ads, for example, have much less social utility than medical research. Second, in real-life context where we gather individuals’ preferred epsilons, they may not perfectly follow a defined distribution. In that case we would need to sample from a distribution. However, we anticipate that this would still enable the selection of the epsilon through our model.

### 3 Adjusting for bias due to $\varepsilon$

Importantly, even a  $\varepsilon$  that is chosen optimally according to our social welfare function may be predictably biased. Any study that samples solely from individuals who are content with higher  $\varepsilon$  and lower privacy can suffer from sample selection bias. Here, we explore two methods for debiasing the estimator. The first involves sampling, while the second involves additions to our welfare function. In both cases, we will focus on the case where the statistic is a mean, or is made DP with the addition of  $\text{Lap}(2s/\varepsilon)$  noise.

#### 3.1 Re-sampling

Assume that the true value of a statistic is  $S$ , that  $\bar{S} = S + \text{Lap}(s/\varepsilon^*)$ , and that  $S$  is calculated using some information about vulnerability  $V_i$ . As before, individual preferences for  $\varepsilon$  follow some distribution; individuals with higher  $\varepsilon_i < \varepsilon$  will not participate; individuals with  $\varepsilon_i > \varepsilon$  will. Finally,  $V_i$ , which relates to our statistic, is correlated with  $\varepsilon_i$ , which we know (although we do not know the precise details of the correlation).

1. Estimate  $\hat{S} = S' + \text{Lap}(2s/\varepsilon)$ , where  $S'$  is calculated using all study participants such that  $\varepsilon_i \geq \varepsilon$ . Thus, we are using half of our privacy budget to calculate a statistic using the data we have.

2. Estimate  $\beta$ , the coefficient that determines the relationship between  $\varepsilon_i$  and  $V_i$  using the study data (i.e. with participants such that  $\varepsilon_i \geq \varepsilon$ ). We do this using differentially private regression with parameter  $\varepsilon/2$ .

3. Extrapolate values of  $\varepsilon_i$  over our distribution of nonparticipants to calculate  $S_{unobserved}$ . This can be done either analytically (e.g. take the integral of  $f(\varepsilon_i)$  over all  $\varepsilon_i < \varepsilon$  for some  $f$  that calculates information) or via sampling, (e.g. sample 2000 possible participants from the unobserved section of the data to calculate  $S_{unobserved}$ ).

4. Combine  $S_{unobserved}$  and  $\hat{S}$ . In the case of a mean, this would mean weighting them equally:

$$\tilde{S} = \int_{\varepsilon_a}^{\varepsilon} f(x)dx(\hat{S}) + \int_{\varepsilon}^{\varepsilon_b} f(x)dx(S_{unobserved}), \quad (9)$$

where  $f(x)$  represents the probability distribution of participants. In the case of a Uniform distribution, this would mean taking a weighted average.

To test the method, we conduct several simulations. We set  $\varepsilon_a = 0.1$ ,  $\varepsilon_b = 1$ , and calculate the value of both the “truncated mean” (i.e. estimated with only study participants) and the “adjusted mean” (i.e. calculated using the procedure outlined above). We compare the bias of these two means with respect to the true mean for  $\varepsilon$  for  $\varepsilon$  at intervals of 0.01 in  $[\varepsilon_a, \varepsilon_b]$ . In each case, the number of study participants is estimated to be  $n = 1000$ , and we conduct 1000 trials.

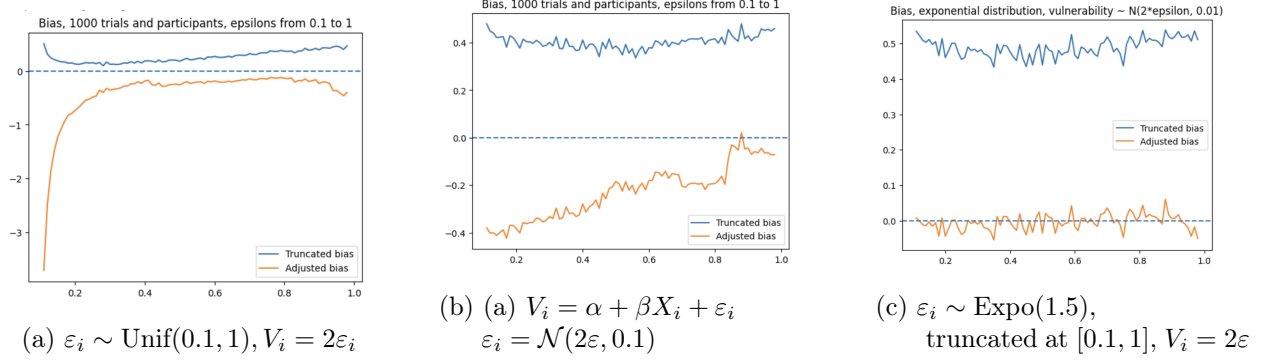


Figure 2: Our method (orange) vs. means based on truncated data (blue).

Importantly, this method assumes that the distribution of preferences for  $\varepsilon_i$  is known, and that the correlation between  $\varepsilon_i$  and  $V_i$  is persistent outside  $[\varepsilon, \varepsilon_b]$ . It is not reliant on the assumption that we have correctly optimized our social welfare function, which in theory means that we might add it to a study with any  $\varepsilon$ , including ones that are poorly chosen.

### 3.2 Welfare functions with bias

Our second method for debiasing our estimator involves adding a term that accounts for bias to our welfare function. We outline a welfare function for calculating a mean statistic here. We adapt the function to:

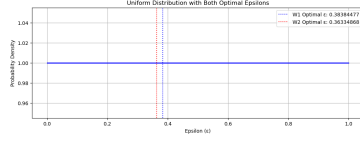
$$W = N \int_{\varepsilon_a}^{\varepsilon_b} \left( \left( \frac{1}{x} \right) \left( 1 - \frac{1}{n\varepsilon} \right) + c \Omega \right) \int_{\max(\varepsilon, \varepsilon_a)}^{\varepsilon_b} \frac{1}{y} dy dx + \int_{\max(\varepsilon, \varepsilon_a)}^{\varepsilon_b} \left( 1 - \frac{\varepsilon}{x} \right) dx \quad (10)$$

where

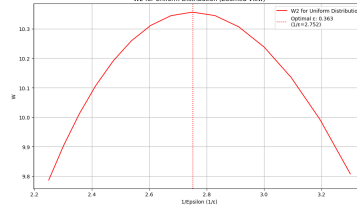
$$\Omega = \left[ \frac{\int_{\varepsilon_a}^{\varepsilon_b} \frac{1}{z} dz}{\varepsilon_b - \varepsilon_a} - \frac{\int_{\varepsilon}^{\varepsilon_b} \frac{1}{z} dz}{\varepsilon_b - \varepsilon} \right]$$

Here,  $c$  represents a constant that means the importance of the bias in the welfare function. In the simulations we set  $c = 0.1$  and set the other constants to the same values as in the subsection 2.4 In the simulations, similarly as in the case of the standard welfare function we account for the PDF of the specific distribution (uniform, normal, mixture of normals, exponential and reversed exponential). In all cases we use the same parameters and weights.

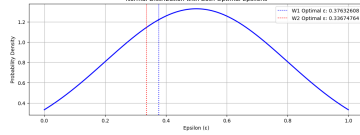




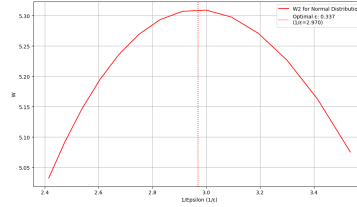
(a) Uniform: Epsilon Distribution



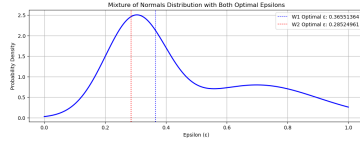
(b) Uniform: Welfare Function



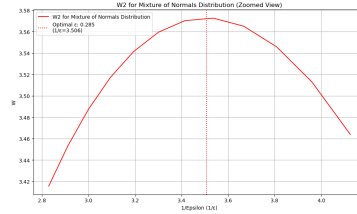
(c) Normal: Epsilon Distribution



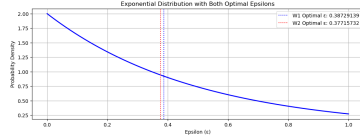
(d) Normal: Welfare Function



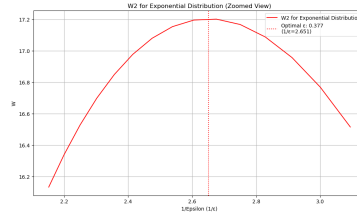
(e) Mixture: Epsilon Distribution



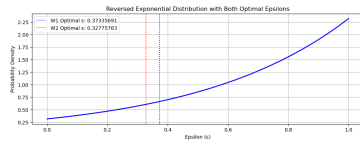
(f) Mixture: Welfare Function



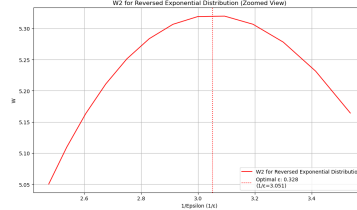
(g) Exponential: Epsilon Distribution



(h) Exponential: Welfare Function



(i) Reversed Exp: Epsilon Distribution



(j) Reversed Exp: Welfare Function

Figure 3: Epsilon distributions and welfare functions with bias across different population models.

We get the following results for distributions - for uniform:  $\varepsilon = 0.363$ , for normal:  $\varepsilon = 0.336$ , for mixture of normals:  $\varepsilon = 0.285$ , for exponential:  $\varepsilon = 0.378$ , for reversed exponential:  $\varepsilon = 0.328$ . Here in this case we can note that all  $\varepsilon$  are smaller from the case of standard distribution, with differences ranging from 0.01 to 0.08.

In the case of the welfare function that accounts for the bias, we can also note the promising results from the simulation. We can note that in all cases the optimum  $\varepsilon$  was lower when we accounted for bias. This aligns with the theoretical intuition - as we use the bias to debias the estimator to include individuals that prefer lower  $\varepsilon$ . Including the bias term in result lead to

the optimum  $\varepsilon$  being lower than in the standard case.

We can note that the differences between optimum  $\varepsilon$  values differ among distributions - which again signals that our model appropriately responds to different preferences of individuals.

## 4 Technical tool

We have created a preliminary tool for both users (prospective study participants) and researchers to understand our work and how it applies to their perspective. Our user section provides an intuitive visual for how different values of epsilon affect their disclosure risk in a given study, while our researcher section aims to help study designers optimize epsilon according to the parameters they set.

Our technical stack includes HTML, CSS, and Python with Flask. We reached out to a few potential users for an informal user study of each section, and have included the results below.

The tool is deployed at [epsilon.dp.pythonanywhere.com](https://epsilon.dp.pythonanywhere.com) – it is accessible online for anyone to use. The code can be found at [github.com/jnair37/cs208-tool](https://github.com/jnair37/cs208-tool).

### 4.1 Design choices

When developing the tool, we prioritized simplicity, clarity, and ease of use as well as applicability to a range of audiences. We included several types of explanatory guides, particularly on the user side, to ensure understandability for general audiences.

### 4.2 Epsilon Visualizer - For Users

The user tool aims to help a user understand the amount of privacy they trade away (their risk of information disclosure) by participating in a given study with a specified  $\varepsilon$ . To calculate a worst-case upper bound for disclosure risk, we use Mehner’s 2021 follow-up Mehner, Voigt, and Tschorsch 2021 to simplify Lee and Clifton’s formula Lee and Clifton 2011 for disclosure risk (given a query about an answer, odds that you will be identified as belonging to a subset who answered a given way). The goal is that given a study with a differential privacy of  $\varepsilon$ , users can make an informed decision about whether or not they want to participate in this study after using our tool. As for a use case relative to a specific study, we assume privacy-aware researchers will want to use it to help users make informed decisions about whether to opt in or out, making researchers’ assumptions on the opt-in/out behavior of users more accurate.

Currently, the tool consists of an interactive slider and graph comparing privacy disclosure risk to epsilon for some set value of  $n$ . This allows users to get a sense of their true disclosure risk. It also includes explanations of what epsilon is on an intuitive level and why this is important for a user to understand before participating. Finally, an "Advanced" section details the formula used and explains the logic at a slightly higher level.

### 4.3 Epsilon Optimizer - For Researchers

The researcher tool aims to help researchers understand how to optimize their epsilon according to our theory. It packages our optimization strategy from above into an interactive format to maximize the welfare function given the  $\varepsilon_a$ ,  $\varepsilon_b$ , and number of participants  $n$ , that the researcher sets. It also visualizes how the total welfare function of the study changes with different epsilon (within the

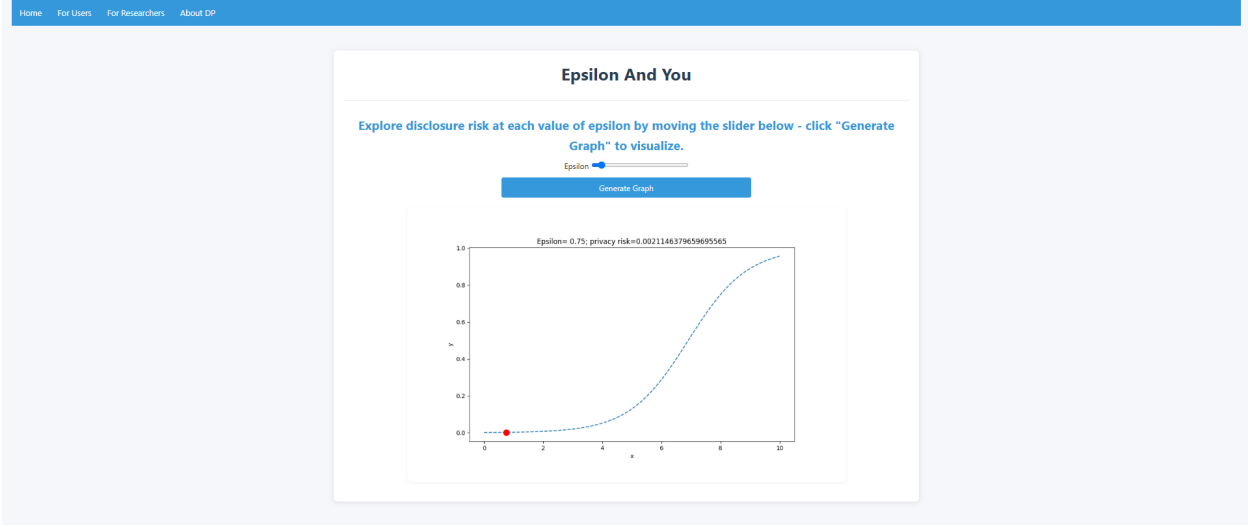


Figure 4: Epsilon Visualizer for Users with  $n=1000$

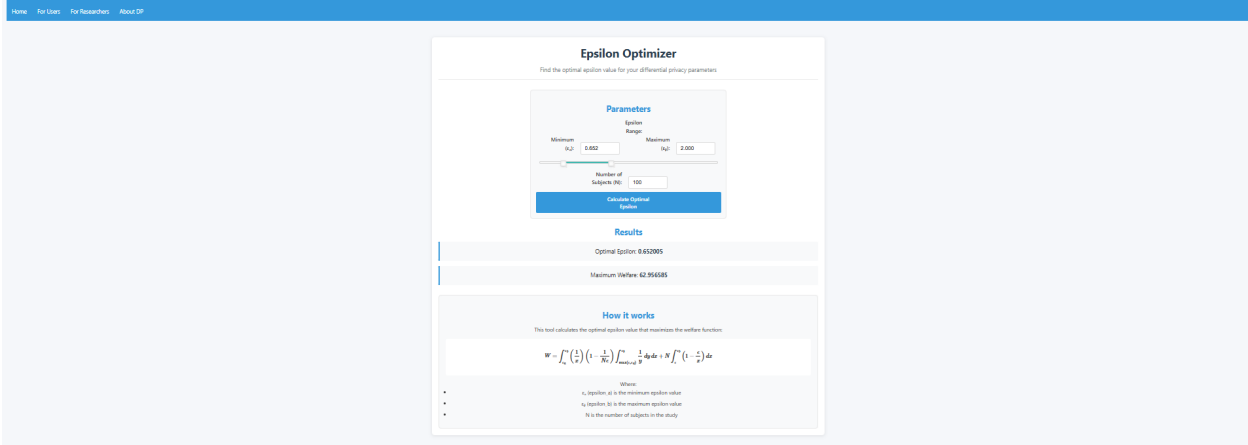


Figure 5: Epsilon Optimizer for Researchers

range of  $\varepsilon_a$  to  $\varepsilon_b$ ) and number of participants. The exact value of the welfare function is not as important as the welfare relative to the maximum we find.

#### 4.4 User Study and Feedback

When developing the tool initially, we conducted a preliminary user study of three participants. The user study was mainly intended for us to get QA-type feedback on the tool during development; if we were to continue working on this in the future and wanted to publish user feedback, we would conduct a full IRB-compliant and truly random study. The three users, who had varying levels of quantitative experience and little to no DP familiarity, understood some aspects of the tool and needed further explanation with others, prompting several improvements. For one, we added a visualization of the welfare curve to the researchers section. For another, we adapted the explanations in the user section to be shorter and more readable, and those in the researcher section to include more detail as well as a visualization. The users also caught minor stylistic issues. At the

same time, some of the broader points raised (one user raised that distrust of the curator was far more concerning than distrust of statistical releases) helped us understand that there are general limitations of DP and it may not be able to reassure users in every potential case.

## 5 Conclusion

We conclude that this theoretical analysis, experimental work, and interactive tool can help both users and researchers make informed decisions surrounding  $\varepsilon$  and their preferred levels of privacy and utility. Importantly, the model and corresponding tools are sensitive to whether we add bias in our welfare function and the distribution of preferences over  $\varepsilon$ , indicating that we would choose different  $\varepsilon$  under different circumstances. We also notice that we often find a  $\varepsilon$  of approximately 0.3 – 0.4, making this a seemingly suitable rule of thumb. Future work might include develop additional methods to quantify the relationship between vulnerability and privacy preference by accounting for other influencing factors and by modeling their interaction using various distributional frameworks. It might also include incorporating new user testing to the tool, updating it so that researchers can choose different distributions for  $\varepsilon_i$ , and developing mechanisms for researchers to estimate the true distribution of  $\varepsilon_i$ . We hope that our contributions to considering optimal  $\varepsilon$  in a world with high privacy-vulnerability correlation are valuable.

## Sources

- Abowd, John M. et al. (2022). “The 2020 Census Disclosure Avoidance System TopDown Algorithm”. In: *Harvard Data Science Review* Special Issue 2. DOI: 10.1162/99608f92.529e3cb9.
- Dwork, Cynthia (2011). “A Firm Foundation for Private Data Analysis”. In: *Communication of the ACM* / vol. 54. URL: <https://dl.acm.org/doi/pdf/10.1145/1866739.1866758>.
- Ghosh, Arpita and Aaron Roth (2010). “Selling Privacy at Auction”. In: *arXiv preprint arXiv:1011.1375*. URL: <https://arxiv.org/abs/1011.1375>.
- Hsu, Justin et al. (2014). “Differential Privacy: An Economic Method for Choosing Epsilon”. In: *arXiv preprint arXiv:1402.3329*. URL: <https://arxiv.org/abs/1402.3329>.
- Krehbiel, Sara (2019). “Choosing Epsilon for Privacy as a Service”. In: *Proceedings on Privacy Enhancing Technologies*. URL: <https://petsymposium.org/popets/2019/popets-2019-0011.pdf>.
- Lee, Jaewoo and Chris Clifton (2011). “How Much Is Enough? Choosing  $\epsilon$  for Differential Privacy”. In: *Information Security*. Ed. by Xuejia Lai, Jianying Zhou, and Hui Li. Berlin, Heidelberg: Springer Berlin Heidelberg, pp. 325–340. ISBN: 978-3-642-24861-0.
- Lin, Yue and Ningchuan Xiao (2024). “Exploring the Tradeoff Between Privacy and Utility of Complete-count Census Data Using a Multiobjective Optimization Approach”. In: *Geographical Analysis* 56.1, pp. 3–25. DOI: 10.1111/gean.12388. URL: <https://onlinelibrary.wiley.com/doi/10.1111/gean.12388>.
- Mehner, Luise, Saskia Nuñez von Voigt, and Florian Tschorsch (2021). *Towards Explaining Epsilon: A Worst-Case Study of Differential Privacy Risks*. Cryptology ePrint Archive, Paper 2021/1125. URL: <https://eprint.iacr.org/2021/1125>.